

Replicative senescence lacks a reproducible enhancer programme, while oncogene-induced senescence is learnable from sequence

Gabriel Kim

Stanford University, Stanford, CA, USA

Correspondence: gabeykim@stanford.edu

Abstract

Designing compact cis-regulatory elements that restrict therapeutic payload expression to senescent cells requires that senescence be a single, learnable regulatory state. We tested that assumption directly against public chromatin data.

Measuring the per-region H3K27ac senescence response and correlating it across independent datasets, we find that two oncogene-induced senescence studies from different laboratories agree at Spearman +0.653 ($n = 88,530$ regions) despite differing in cell line, tissue, induction construct and processing pipeline. Three within-replicative comparisons give +0.054, +0.144 and +0.268, the first between two subseries of one deposition sharing laboratory, pipeline and genome build. Two of the three fall inside the cross-mechanism range of -0.164 to +0.164, and none approaches the oncogene-induced ceiling. The measurements do detect senescence: CDKN2A promoter-proximal H3K27ac falls in every arm examined, and significantly so within six of seven by anchor-level Wilcoxon test, including each replicative arm individually.

Cross-mechanism divergence survives a control removing laboratory, cell line, pipeline and genome build simultaneously (+0.114), and reproduces at the transcriptome and model levels. Comparing multiple senescence conditions against one shared control inflates apparent agreement, with an observed +0.564 against a constructed null of +0.523.

Training a prediction head on a frozen Borzoi trunk using one laboratory's oncogene-induced senescence data predicts another's at Spearman +0.5484 — 86.1% of the data's own reproducibility ceiling — and AUC 0.9070, with 0 of 200 shuffled-label permutations beating it. Classification, however, does not imply design readiness: the strongest available motif lever reaches only 0.773 of the gain required to move a sequence to the 95th percentile, and in three of four motif families the predicted effect of synthetic insertion disagrees with observed genomic enrichment.

All datasets provide $n = 2$ per arm, which bounds these conclusions. For computational work on senescence regulation, the mechanism must be selected before the design target, and published replicative-senescence enhancer landscapes should be treated as dataset-specific until reproduced at higher replication.

Keywords: cellular senescence, enhancer, H3K27ac, reproducibility, cis-regulatory element design, sequence-to-activity models, oncogene-induced senescence.

1. Introduction

Cellular senescence is a stable cell-cycle arrest adopted in response to telomere attrition, oncogene activation, DNA damage and other stresses. Senescent cells accumulate with age and secrete inflammatory factors, which makes them an attractive target for therapies acting selectively on the senescent state.

Selectivity is the obstacle. Gene therapy can deliver a payload but cannot easily restrict where that payload is expressed, and a payload that kills or silences is dangerous wherever it leaks. Compact cis-regulatory elements that restrict expression to a defined cell state are one route to that control, and they must be compact:

adeno-associated virus carries roughly 4.7 kb in total, so every base spent on regulation is a base unavailable to the therapeutic gene.

Deep learning models that predict regulatory activity from DNA sequence have made such elements designable (Gosai et al. 2024; Taskiran et al. 2024; DaSilva et al. 2026). Nearly all of this work targets cell types; the few cell-state precedents concern T-cell exhaustion (Zhang et al. 2025; Wu et al. 2019).

Applying the same approach to senescence requires an assumption that has not been tested: that senescence is one regulatory state. The field has established that senescence is phenotypically heterogeneous, that no universal marker exists, and that senescent cells can be classified into distinct senotypes by input and function (Schafer et al. 2026). Transcriptomic comparisons across inducers report limited gene-level overlap with conserved pathway-level enrichment (Casella et al. 2019; Bridge et al. 2026). What has not been established is whether senescence induced by different mechanisms shares a common cis-regulatory programme, measured quantitatively and calibrated against what agreement looks like when the mechanism is shared.

Three findings follow. Replicative senescence, the canonical Hayflick model on which much of the field's data rests, does not produce a reproducible enhancer-level response across independent datasets. Oncogene-induced senescence does, and a sequence model trained on one laboratory's OIS data predicts another's. And model-guided design of such elements nevertheless hits a measurable ceiling.

2. Methods

2.1 Data

Every accession was verified against a live NCBI GEO, BioProject or ENCODE record before use; 44 were verified in total. Verification was by live query against the repository record rather than by citation.

Accession	Cells	Mechanism	Assays	n per arm
GSE175533	WI-38 lung fibroblasts	Replicative, PDL20-53	ATAC, RNA	2
GSE146585	BJ foreskin fibroblasts	Replicative	H3K27ac, H3K4me2, ATAC, RNA, Hi-C	2
GSE106146	WI-38 / IMR90	Replicative + irradiation	H3K27ac, H3K4me1, p300, CBP	2
PRJNA439280	WI-38	Replicative, RAS-OIS, RAF-OIS	ATAC, H3K27ac, H3K4me1/3	2
GSE206402 / GSE205898	GM21 skin fibroblasts	OIS, ER:RAS	ATAC, H3K27ac, H3K4me1	2
GSE74238	IMR90	OIS, HRasV12	H3K27ac, BRD4	2

Source publications, in table order: Chan et al. (2022) for GSE175533; Suter, Friedman et al. (2025) for GSE146585; Sen et al. (2019) for GSE106146; Martínez-Zamudio et al. (2020) for PRJNA439280; Martínez-Zamudio et al. (2023) for GSE206402 and GSE205898; and Tasdemir et al. (2016) for GSE74238. GSE210285 was excluded: it deposits four bigWig files with no peak calls, and the ChIP, RNA and OIS arms described

in its source publication are absent from the accession. PRJNA439280 was excluded from model training as raw reads only.

2.2 Region harmonization

Genome build was confirmed per file rather than taken from series headers, which frequently omit it. Where liftover was required, hg19ToHg38.over.chain.gz was applied to region midpoints with fixed-width recentring, so only midpoints required lifting. For the OIS training set, 99.98% lifted with 21 failures; no span-integrity check was run on that build. The enhancer cross-mechanism build achieved 99.92% midpoint success with 99.96% span preservation, and the replicative hg18 build 99.87%.

Consensus H3K27ac regions were summit-clustered and required in at least 2 of 4 GM21 samples, yielding 112,099 regions. 111,671 survived into the training object; losses were 247 regions whose model window ran past a chromosome end, 154 before a start, 21 liftover failures and 6 ambiguous lifts — 428 regions, 0.3818%.

No dataset was reprocessed from raw reads. All analyses read deposited bigWig signal, and for GSE175533 a deposited log2 fold change that was not requantified. Where this work describes two arms as sharing a processing pipeline, that refers to the depositing laboratory's own pipeline, evidenced by a common processing suffix in the deposited filenames, rather than to uniform reprocessing from FASTQ performed here. Uniform reprocessing from raw reads would strengthen every comparison reported and is the single most valuable extension of this analysis.

Comparison identifiers used throughout (R1-R3, A1-A2, X1-X4, and a-d) map to specific accession pairs with their cell lines, originating laboratories and genome builds recorded; the complete mapping accompanies the deposition.

Two coordinate systems are stored per region. H3K27ac marks nucleosomes flanking a regulatory element and is depleted over the nucleosome-free region where transcription factors bind; the median distance from H3K27ac summit to nearest ATAC summit is 245 bp. Measurement and training use the H3K27ac summit; sequence-design analyses use the ATAC summit. A 200 bp window centred on the H3K27ac summit would fall in the acetylation trough.

2.3 Model and caching

Borzoi (Genentech/borzoi-model, human_rep0 checkpoint, 185,892,699 parameters, 524,288 bp input, 32 bp bins, 7,611 output tracks, hg38) (Linder et al. 2025) was loaded through gReLU 1.1.0 (Lal et al. 2025). TF32 was disabled on both cudnn and matmul paths. The prediction head used a linear activation and was verified empirically to emit negative predictions, since targets are signed.

Borzoi predicts a 196,608 bp span per input and enhancers cluster, so 111,671 regions were covered by 8,459 distinct windows — 13.201 regions per window, by greedy interval covering per chromosome under a width-aware criterion requiring the full 1 kb region to fit inside the span, which reduces the usable span to 195,608 bp. A point-covering criterion on the same regions gives 8,432 windows; the executed width-aware value is reported.

Trunk embeddings were mean-pooled over each region's bins. The pooling identity is exact rather than approximate: the head applies a 1x1 convolution followed by average pooling, both linear, so they commute. Worst observed relative deviation was 3.31e-07 against a 1e-04 tolerance.

Two preflight checks aborted the run on failure. Bin arithmetic was verified by confirming that regions sharing a window resolve to bin ranges consistent with their genomic separation: 5,387 adjacent pairs checked,

0 inconsistent. On the reverse-complement pass output bins reverse, so a region at forward bins [b0, b1) occupies [6144-b1, 6144-b0); this was implemented as explicit index arithmetic rather than by re-extracting sequence.

RefSeq GRCh38 contains IUPAC ambiguity codes that survive uppercasing and are rejected by gReLU's allowed-base set. Any base outside {A,C,G,T,N} was mapped to N, with affected windows and characters counted and recorded.

2.4 Splitting

The head was trained on GM21 and evaluated on IMR90 — a cross-study, cross-tissue, cross-laboratory holdout. chr9 and chr6 were held out from every stage.

Early stopping used a GM21-internal chromosome holdout (chr8, chr16, chr20). Stopping on IMR90 would have selected the checkpoint using the evaluation study.

No pooled batch correction was applied. Study and validation fold are perfectly confounded by design, so a correction fitted across both would use evaluation information at training time. Targets are per-study log2 fold changes, within-study contrasts in which the study offset cancels by construction.

2.5 Metrics and controls

AUC thresholds were pre-registered before results were viewed: a region counts as positive at measured log2FC of +1 or above and negative at -1 or below, with intermediate regions excluded as ambiguous. Evaluable n and ambiguous count accompany every AUC.

Model performance is additionally reported as a fraction of the measured cross-dataset reproducibility ceiling, since a model cannot exceed the reproducibility of the quantity it predicts.

Shuffled-label permutations used 200 draws minimum, following the standard permutation treatment of classifier performance (Ojala and Garriga 2010). The minimum attainable empirical p at 200 permutations is $1/201 = 0.00498$, and we state this floor wherever it is reached, since permutation p-values are bounded below by the number of draws (Phipson and Smyth 2010). Where a contrast admits few distinct partitions the null was enumerated exhaustively rather than sampled: a 3-versus-3 contrast admits exactly $C(6,3) = 20$ partitions, so additional permutations resample the same 20 values and would report a falsely precise p.

An untrained randomly-initialized head established the floor. A shuffled region-to-embedding assignment tested whether sequence is used at all.

2.6 Design probes

Motif insertion used consensus sequences for NF-kB/RELA, C/EBP-beta, FOS::JUN and ETS inserted into random 200 bp background at increasing copy number, each with a matched scrambled control, at the ATAC-summit coordinate. Combination tests held total inserted copy number constant on an identical slot grid so that mixtures were compared against singles at equal insertion load; because motif lengths differ, mixtures insert fewer bases at equal copy number, biasing conservatively against combinations. Ten random backgrounds per condition.

In-silico saturation mutagenesis was run over 500 bp windows. Position weight matrix enrichment used JASPAR CORE vertebrates matrices RELA MA0107.1, CEBPB MA0466.2, FOS::JUN MA0099.3 and ETS1 MA0098.3. Matrices rather than consensus strings, and a 500 bp rather than 200 bp window, are required for adequate match counts at this region size.

Motif ablation removed native sites from real regions and measured the change in predicted activity, stratified by whether the region was high- or low-response, with NF- κ B as a positive control.

Locus-level tests at CDKN2A use distance to the gene body rather than to a transcription start site, so "promoter-proximal" denotes gene-body-proximal regions. The sign pattern is unchanged at cutoffs of 0, 1, 2, 5, 10 and 20 kb. Promoter-proximal and distal sets are complementary by construction and do not overlap. A sign test across datasets cannot resolve below $p = 0.0625$ at five datasets, the same discreteness limit that applies to the 3-versus-3 permutation null, so within-arm Wilcoxon tests on anchor-level values are reported instead; these use variation the sign test discards.

2.7 Numerical verification

All 72 numerical claims in this manuscript were audited line by line against their source artifacts. Structural consistency was verified: evaluable plus ambiguous equals total in all 8 AUC blocks, region losses sum exactly, and the 200 stored permutation values recount to zero beating the model.

2.8 AI assistance

Data processing, analysis code, audit scripts, and manuscript drafting were carried out with the assistance of an AI system (Claude, Anthropic), operating under task specifications and editorial direction from the author. The author designed the study, made all interpretive decisions, verified every reported number against committed outputs, and is responsible for all claims.

3. Results

3.1 The public senescence chromatin corpus

Forty-four accessions were verified live against NCBI GEO, BioProject or ENCODE records. Six carry the analyses reported here; a verification log recording the method and date for every accession accompanies the deposition.

Mechanism	Usable studies with a non-senescent control arm	Assays
Replicative	4	ATAC, H3K27ac, H3K4me1/2, p300, CBP
Oncogene-induced	3	ATAC, H3K27ac, H3K4me1, BRD4
Therapy-induced	1, inadequate	ATAC, ChIP, no control arm, no replicates
Mouse replicative	1	H3K27ac, H3K4me1, Hi-C

Every human study provides $n = 2$ per arm; none reaches $n = 3$. This bounds every result reported here.

Two absences constrain the corpus. No massively parallel reporter assay or STARR-seq dataset in senescent cells exists publicly, which forecloses training a reporter-native model and requires transfer from accessibility or histone marks. And no therapy-induced senescence chromatin dataset with matched proliferating controls and replicates exists, which matters because therapy-induced senescence is the mechanism most readily produced by contract laboratories and therefore the most natural target for experimental validation.

We excluded GSE210285 (2BS fibroblasts, replicative): it deposits four bigWig files with no peak calls, and the ChIP, RNA and OIS arms described in its source publication are not present under that accession.

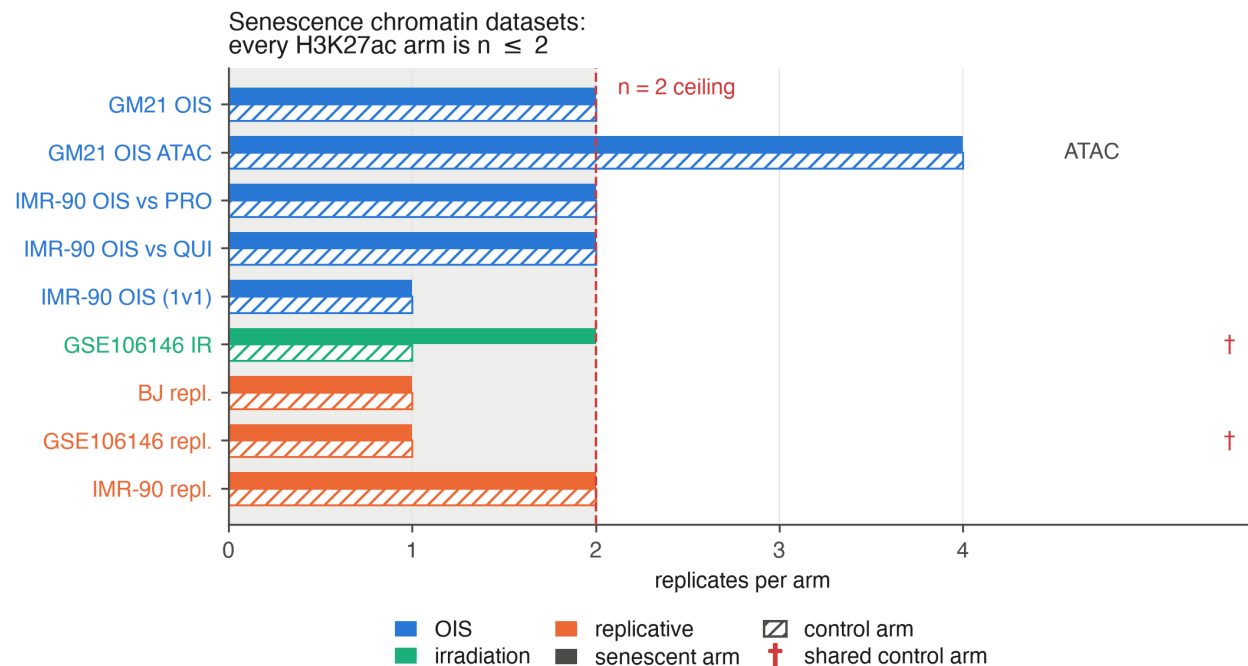


Figure 1. Data landscape. Senescence chromatin datasets by induction mechanism. Filled bars give replicates in the senescent arm, hatched bars the matched non-senescent control; colour encodes mechanism. Shading and the dashed line mark $n = 2$. Every H3K27ac arm in every study is $n \leq 2$ per side; only GM21 ATAC reaches 4 versus 4. GSE106146 contributes two contrasts, replicative and irradiation, that share one proliferating control arm, which is the source of the shared-denominator artifact quantified in Figure S3. WI-38 replicative ATAC is deposited as a log2 fold change only, with no replicate-level data, and is not plotted. This $n = 2$ ceiling bounds every correlation reported. No error bars: counts.

3.2 Oncogene-induced senescence produces a reproducible enhancer response across laboratories

Any claim that a dataset fails to reproduce requires first showing that the method detects reproduction when it occurs. Agreement within one mechanism therefore establishes the reference against which agreement across mechanisms is read, correlating the per-region H3K27ac senescence response between datasets on a harmonized region set (Methods).

Two OIS studies agree at Spearman $+0.653$ ($n = 88,530$ regions). These studies are from different laboratories (Martínez-Zamudio et al. 2023 and Tasdemir et al. 2016, published seven years apart) and differ in cell line and tissue of origin (GM21 skin fibroblasts versus IMR90 lung fibroblasts), induction construct (ER:RAS versus HRasV12), ChIP protocol and processing pipeline. Agreement at this level across that much technical distance establishes the measurement's sensitivity, and it sets up the contrast that follows: two different laboratories measuring oncogene-induced senescence agree at $+0.653$, while a single laboratory measuring replicative senescence agrees with itself at $+0.054$, or $+0.146$ on regions shared between the two comparisons.

The correlation is stable across response-magnitude thresholds ($+0.640$ to $+0.678$), as a coherent programme should be. Swapping the assay within a single study costs little — ATAC against H3K27ac in GM21 gives $+0.607$ ($n = 36,560$) — so assay identity is worth roughly 0.05 of correlation, a reference point for the cross-mechanism comparisons below.

These measurements also behave as senescence measurements should. At CDKN2A, promoter H3K27ac falls in both OIS studies (-1.03, -1.07) while distal regions rise (+0.24, +0.49) — the dissociation expected if regulation at this locus is enhancer-driven rather than promoter-driven.

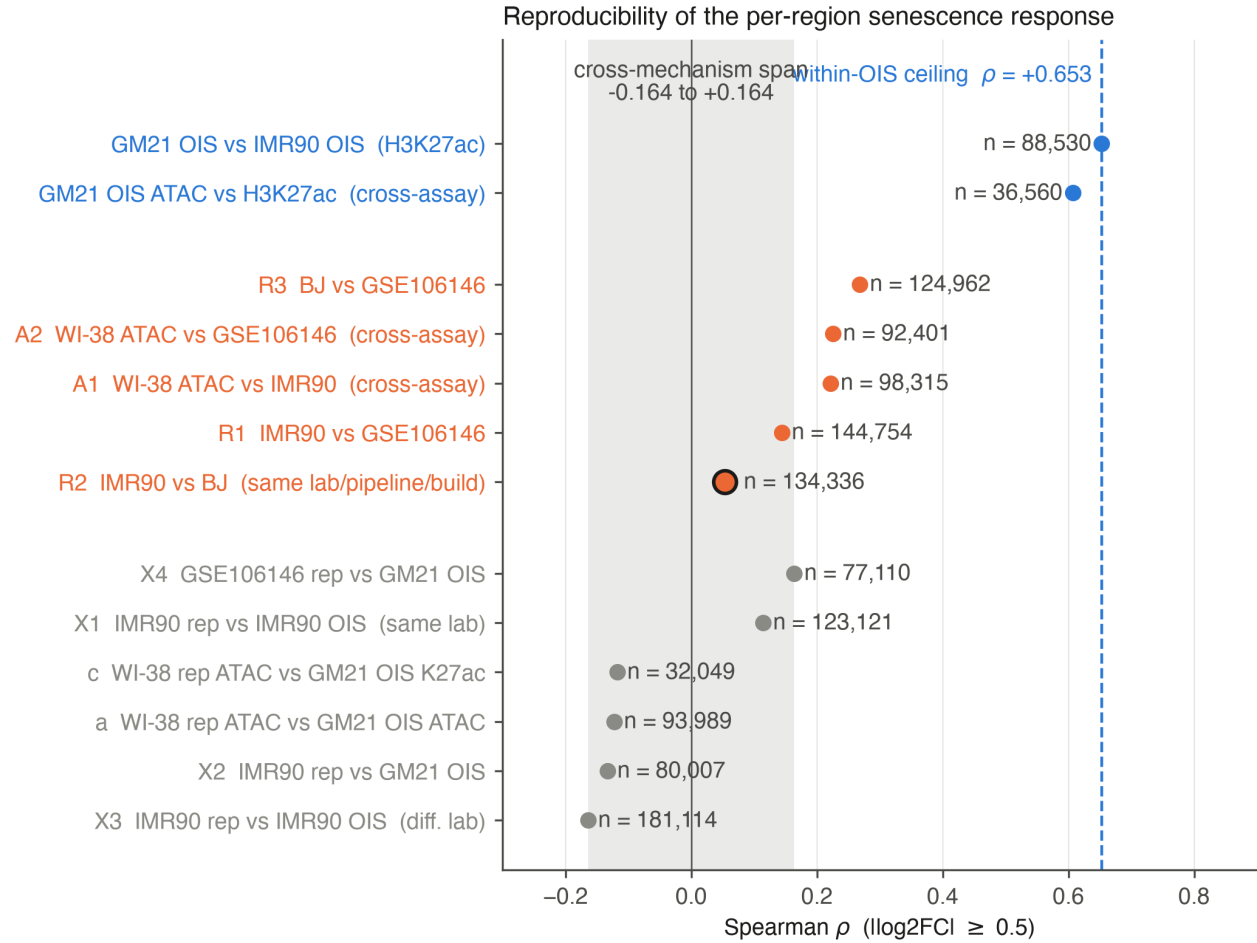


Figure 2. The reproducibility matrix. Per-region Spearman correlation of the senescence response between every pair of datasets, at the primary response-magnitude threshold $|\log_2FC| \geq 0.5$, grouped by whether the two datasets share an induction mechanism. Each point is one comparison; n regions entering that correlation is printed beside it. The dashed blue line marks the within-OIS ceiling, $\rho = +0.653$ (GM21 versus IMR90 H3K27ac, $n = 88,530$). Grey shading spans the four cross-mechanism H3K27ac comparisons, -0.164 to +0.164. The ringed point is R2 (IMR-90 versus BJ replicative, $\rho = +0.054$, $n = 134,336$): the comparison with no technical confound, since both arms are subseries of a single deposition sharing originating laboratory, pipeline and genome build, and therefore the load-bearing value. On the 29,343 regions shared with the within-OIS comparison this value rises to +0.146 while the ceiling falls only to +0.645, so the ordering is independent of region set even though the smallest value is not. Two of the three within-replicative H3K27ac comparisons fall inside the cross-mechanism span; the third sits just above it, as do the two cross-assay within-replicative comparisons A1 and A2. No within-replicative comparison approaches the within-OIS ceiling. No error bars: each point is a single correlation over the stated n .

3.3 Replicative senescence does not reproduce against itself

Applying the same measurement to replicative senescence gives a different answer.

Comparison	Spearman	n regions
Same laboratory, pipeline, aligner and genome build	+0.054	134,336
Cross-laboratory	+0.144	144,754
BJ fibroblasts against WI-38/IMR90	+0.268	124,962
<i>Reference: within-OIS ceiling</i>	<i>+0.653</i>	<i>88,530</i>

The three within-replicative values span +0.054 to +0.268, against a cross-mechanism range of -0.164 to +0.164. Two of the three fall inside that range; the third, at +0.268, exceeds it. Two further within-replicative comparisons made across assays give +0.21 and +0.22, also just above the span. No within-replicative comparison of any kind approaches the +0.653 within-OIS ceiling, and on the comparison carrying no technical confound two datasets of the same mechanism agree no better than two of different mechanisms.

The cross-assay comparisons correlate an ATAC response against an H3K27ac response. The equivalent within-study comparison in OIS reaches +0.607, so the two assays track one another when a shared programme is present, and a low cross-assay value within replicative senescence is therefore informative rather than expected.

The first row is the load-bearing one. Both arms are subseries of one deposition sharing originating laboratory, pipeline and genome build, so it carries no technical confound and heterogeneity cannot account for +0.054. Restricted to the 29,343 regions shared with the within-OIS comparison it rises to +0.146, still far below the +0.645 the same restriction gives the ceiling. Both figures are reported throughout, since the smaller is specific to this comparison's own region set.

It does retain a biological confound. The two arms are different cell lines — IMR-90, foetal lung, against BJ, neonatal foreskin — so cell-line identity is not controlled. Comparison X1 bounds that contribution: it holds laboratory, genome build and cell line constant (IMR-90 throughout) and varies only the induction mechanism, giving +0.114. Within one laboratory and one cell line, replicative senescence agrees better with oncogene-induced senescence than it does with replicative senescence measured in a second cell line by the same laboratory. The cell-line difference in R2 therefore cannot be carrying the result.

The cross-laboratory comparison is strongly threshold-dependent, rising from +0.063 at the loosest magnitude threshold to +0.327 at the strictest, where the OIS ceiling is flat across the same range. Dependence on the cutoff is characteristic of correlation driven by a small number of extreme regions rather than by a coherent programme.

These measurements do detect senescence The result above would be uninteresting if the replicative datasets simply failed to capture the senescent state. They do capture it. CDKN2A promoter-proximal H3K27ac falls in all three replicative datasets (-0.85, -0.63, -0.43), closely matching the two OIS datasets (-1.03, -1.07). Across the seven arms shown in Figure 4 the promoter response is negative in every one, and within-arm Wilcoxon tests on anchor-level values place it significantly below zero in six of seven, including each of the three replicative arms individually ($p = 0.0098, 0.0391, 0.0371$). The exception is a one-versus-one OIS arm whose response is indistinguishable from zero (-0.055, $p = 0.73$). The pattern is insensitive to the distance cutoff: all five primary datasets remain negative at 0, 1, 2, 5, 10 and 20 kb.

Each replicative assay therefore detects the expected biology at the canonical senescence locus. Whatever fails to reproduce between these datasets, it is not that they failed to measure senescence.

The distal response at this locus is less clean than the promoter response. It is positive in both OIS datasets (+0.24, +0.49) and in two of the three replicative datasets (+0.215, +0.311), with only one replicative dataset negative (-0.17). The apparent dissociation therefore rests on a single dataset, and with two OIS against three replicative arms no test can distinguish the groups — Fisher's exact reaches only $p = 0.1$ even under perfect separation. The pattern is therefore descriptive. The reproducibility finding rests on the genome-wide correlations, not on this locus.

3.4 Statistical treatment of the correlation differences

Bootstrap confidence intervals accompany every correlation (1,000 resamples over regions; standard errors 0.0017 to 0.0054). Differences were tested on paired bootstraps, every pair sharing at least 49,000 regions.

Difference tested	n shared	Difference	95% CI	p
Ceiling – R2	56,796	+0.5740	[+0.5651, +0.5832]	<0.001
Ceiling – R1	50,701	+0.5651	[+0.5560, +0.5740]	<0.001
Ceiling – R3	54,195	+0.3255	[+0.3172, +0.3336]	<0.001
R2 – cross-mechanism upper bound	49,818	-0.1059	[-0.1166, -0.0947]	<0.001
R3 – cross-mechanism upper bound	52,164	+0.1551	[+0.1464, +0.1642]	<0.001

No resample crossed zero in any test; p is capped at <0.001 by the 1/2000 resolution floor. The cross-mechanism upper bound of +0.164 is itself an estimate rather than a constant, so these are correlation-against-correlation tests; treating it as a fixed bound gives the same signs and the same significance.

Sensitivity to the region set The comparisons rest on different region sets, from 88,530 regions for the within-OIS ceiling to 181,114 for the widest cross-mechanism comparison. The ceiling therefore sits on the smallest set of any comparison, raising the question of whether it is elevated by that restriction. Recomputing the four core comparisons on their 29,343 shared regions answers it: the ceiling falls by 0.008, from +0.653 to +0.645. It is not an artifact of region selection.

One value does move. R2 rises from +0.054 to +0.146 on the common set, becoming indistinguishable from R1 at +0.141, and the gap between the ceiling and R2 narrows from 0.599 to 0.499. The value +0.054 is therefore specific to R2's own region set, and both figures are given wherever it is quoted. The ordering and its magnitude are unaffected; only the precise value of the smallest number depends on the region set.

The intersection across all sixteen comparisons retains only 1,706 of 230,988 regions (0.7%) and is dominated by the two cross-assay comparisons. Fourteen of sixteen correlations rise on it, reflecting selection toward regions with a large response in every channel rather than robustness; it supports no conclusion. Region loss under intersection is strongly non-random in response magnitude but essentially random across chromosomes.

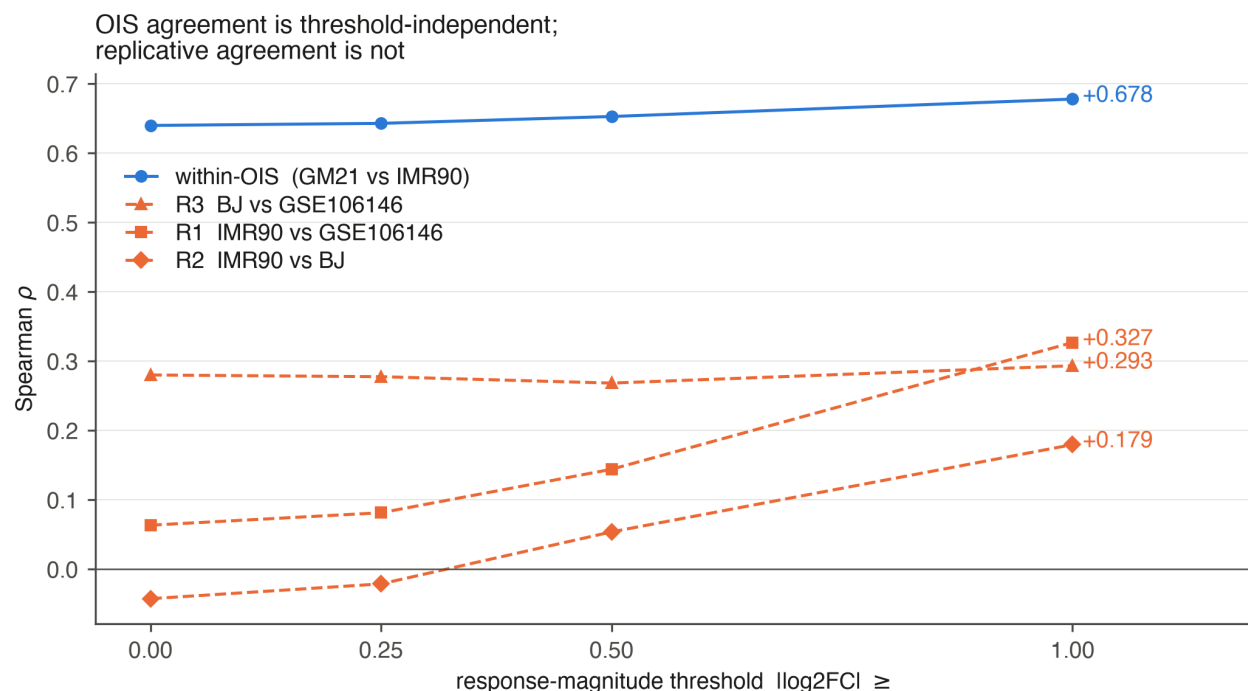


Figure 3. Threshold sensitivity. Spearman correlation as a function of the response-magnitude threshold applied before correlating, one line per comparison. The within-OIS ceiling is flat across thresholds (+0.640 to +0.678); the cross-laboratory within-replicative comparison R1 rises steeply (+0.063 to +0.327). Flatness indicates a correlation carried by the bulk of regions; steep threshold dependence indicates one carried by a small number of extreme regions. No error bars: point estimates.

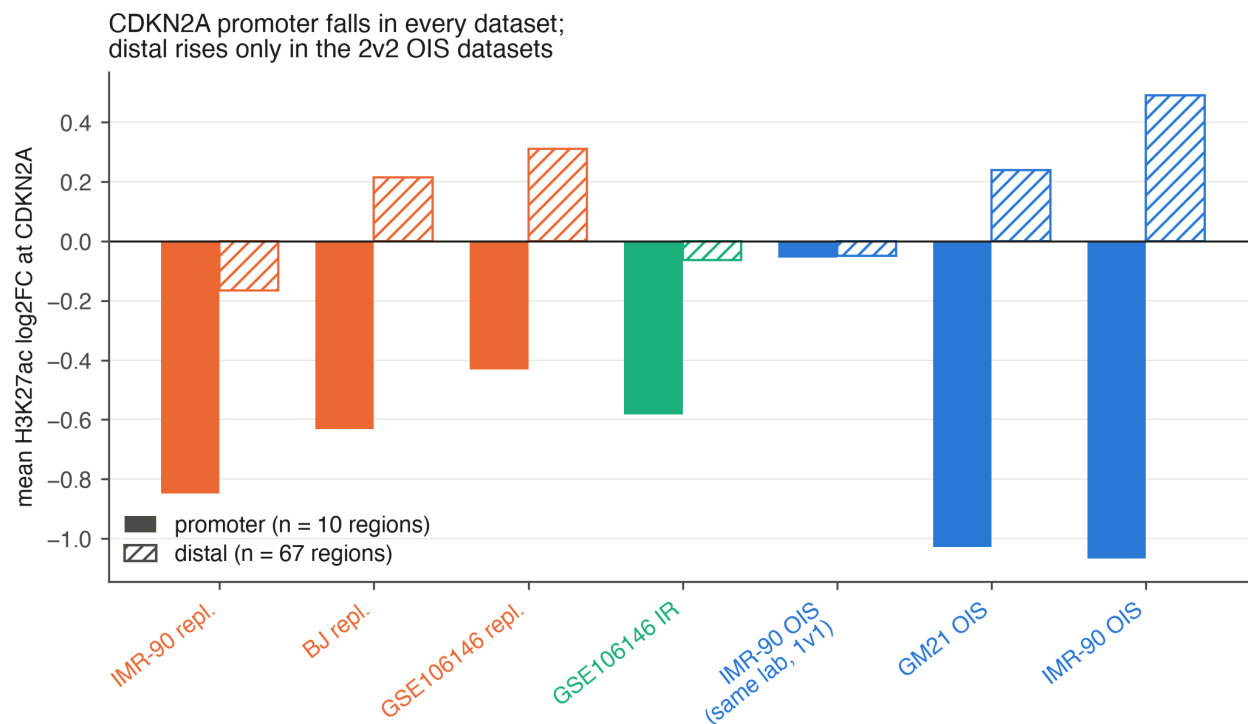


Figure 4. CDKN2A promoter versus distal response. Mean H3K27ac senescence response at the CDKN2A locus, split by promoter-proximal ($n = 10$ regions) and distal ($n = 67$ regions), across seven arms drawn from

four studies. The five datasets referred to in the text are the three replicative arms and the two two-versus-two OIS arms; the irradiation arm and a one-versus-one OIS arm are shown here for completeness and enter no correlation reported in the main text. The irradiation arm of GSE106146 and a one-versus-one IMR-90 OIS arm are shown for completeness and are not used in any correlation reported in the main text. The promoter response is negative in every dataset, replicative and oncogene-induced alike, establishing that each measurement detects the senescent state. The distal response is positive in both OIS datasets but inconsistent in sign across the three replicative datasets. This dissociation separates measurement failure from genuine non-reproducibility of the distal programme. No error bars: means over the stated regions.

3.5 Senescence mechanisms do not share an enhancer programme

Cross-mechanism per-region response correlations span -0.164 to +0.164 across four comparisons on 80,000 to 181,000 regions each. Against the within-OIS ceiling of +0.653, swapping the mechanism costs roughly 0.5 to 0.8 of correlation, where swapping the assay costs roughly 0.05. Two of the four cross-mechanism comparisons are negative and two are positive, so the mechanism difference abolishes agreement rather than reversing it systematically.

The controlled comparison is decisive: holding cell line, pipeline and genome build constant and varying only mechanism gives +0.114, so divergence is not an artifact of comparing across laboratories.

The finding reproduces in a second modality with a different failure profile. At the gene-expression level, cross-study Spearman of the per-gene senescent-versus-proliferating response is +0.531 between two OIS studies but +0.142 and +0.093 between replicative and either OIS study.

It reproduces a third way at the model level. A single prediction head trained on pooled data from both mechanisms reaches AUC 0.7843 when its output axis is mechanism-specific ($p = 0.00498$, 0 of 200 permutations beating it), but 0.6895 and failing its shuffled-label control (27 of 200, $p = 0.139$) when the same head is asked to predict a unified senescent-versus-proliferating contrast. Pooling the data helps; pooling the contrast does not.

Irradiation-induced senescence appears to constitute a third programme resembling neither, sitting at +0.115 with replicative and +0.083 with OIS. This rests on a single irradiation dataset, so it cannot be distinguished from dataset-specific variation in the way the replicative and OIS claims can, and is suggestive rather than established.

A shared control inflates apparent agreement

The irradiation result required care. Comparing the irradiation and replicative responses within a single study gives +0.564 — the highest within-study-like value we measured, and one that would have supported a conclusion of shared programme.

Both responses are computed against the same proliferating control. We therefore constructed a null in which two numerators with no shared biology are divided by that same denominator; it returns +0.523. The observed value is barely above the artifact floor.

This is the classical spurious correlation of ratios sharing a common component, described by Pearson (1897), whose own calculations produced correlations near 0.5 from uncorrelated measures, and quantified by Monte Carlo in Dunlap et al. (1997). It arises in any senescence study comparing multiple induction conditions against one control, and that design is common.

With independent controls, the same comparison gives +0.115.

3.6 A sequence model predicts the OIS enhancer response across laboratories

Because OIS reproduces and replicative does not, OIS is the only mechanism in this corpus against which a sequence-to-activity model can be meaningfully trained and tested.

A small prediction head was attached to a frozen Borzoi trunk, trained on the GM21 skin-fibroblast H3K27ac senescence response, and evaluated on IMR90 lung fibroblast data from a different laboratory. Chromosomes 9 and 6 were held out from every stage so that CDKN2A and CDKN1A remain clean benchmark loci. Early stopping used a GM21-internal chromosome holdout; stopping on IMR90 would have selected the checkpoint using the evaluation study.

Metric	Held-out IMR90	chr9 + chr6
Spearman	+0.5484	+0.5250
Fraction of reproducibility ceiling	86.1%	82.4%
AUC	0.9070	0.8899
Evaluable regions	26,936 of 100,070	3,267 of 11,601

Performance is given as a fraction of the measured cross-dataset reproducibility ceiling (+0.637 on this harmonized region set) because a model cannot exceed the reproducibility of the quantity it predicts. At 86.1% of that ceiling, the remaining headroom is small.

The senescent-versus-quiescent contrast in IMR90 gives +0.5895 and AUC 0.9354, so the model is not merely detecting growth arrest.

Controls. No permutation of 200 shuffled-label draws beat the model on either Spearman or AUC, giving an empirical p of 0.00498 — the resolution floor of a 200-permutation test. An untrained randomly-initialized head gives -0.0903 and AUC 0.3842. Shuffling the assignment between regions and their sequence embeddings collapses performance from +0.548 to +0.025, establishing that the model uses sequence rather than per-region marginals. Predictions are not collapsed: prediction standard deviation 0.5137 against a measured 0.9601.

On held-out chr9, the model reproduces the CDKN2A signature in direction: predicted -0.594 at the promoter against a measured -1.341, and predicted +0.198 distally against a measured +0.421.

All of the above rests on $n = 2$ per arm in both studies.

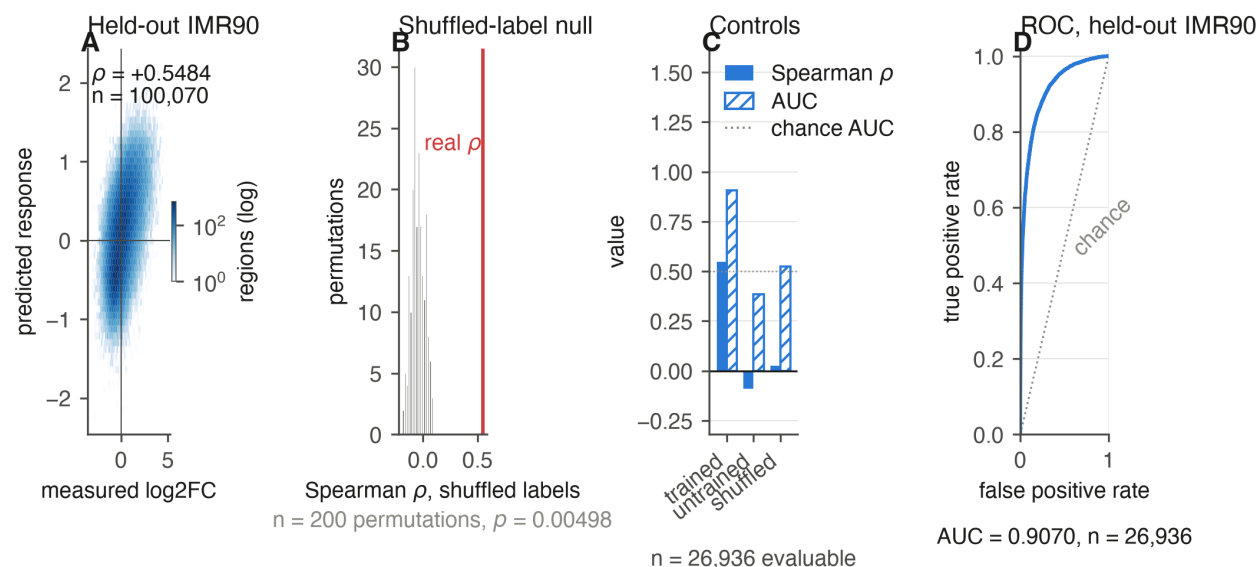


Figure 5. OIS model performance on a cross-laboratory holdout. The model is trained on GM21 skin fibroblast H3K27ac senescence response and evaluated on IMR90 lung fibroblast data from a different laboratory. Panels show predicted against measured per-region response with the reproducibility ceiling marked; the 200-permutation shuffled-label null with the real value indicated, where the resolution floor is $p = 1/201 = 0.00498$; the control comparison against an untrained head and a shuffled region-to-embedding assignment; and the ROC curve with evaluable n stated. The AUC null is reported as summary statistics rather than a distribution because per-permutation AUC values were not written at run time. $n = 2$ per arm in both studies. No error bars: distributions and point estimates.

3.7 Model-guided design hits a measured ceiling

A model that classifies well does not necessarily guide design. We tested whether the learned signal is strong enough to optimize against.

Moving a sequence from the median to the 95th percentile of predicted activity requires a gain of +0.861. The best achievable gain from motif insertion is +0.6657 (SEM 0.0487, SD 0.154, $n = 10$ random backgrounds), from NF-kB and ETS motifs clustered at 12 copies — a shortfall of +0.196, reaching 0.773 of requirement.

Three observations bound why.

Combining motif families adds nothing. The best pair against the best single family at matched insertion load, paired on identical backgrounds, differs by +0.0639 (SEM 0.0414, SD 0.131, paired t $p = 0.157$). The result is robust to test choice: Wilcoxon $p = 0.193$, sign test $p = 0.344$, 95% CI [-0.030, +0.158]. The t -test is the most favourable of the three to the combination. Compositional diversity is not the limiting ingredient.

Dose reverses rather than plateaus. Four of five conditions peak at 12 copies and ETS peaks at 6; all five sit below their peak at 20 copies, where the cassette is entirely motif. Marginal gain per copy becomes negative beyond the peak. Saturation in motif multiplicity has been reported in cell-type enhancer design; reversal is a stronger constraint than saturation, because it means additional motif content actively costs predicted activity.

Spacing matters more than composition. Clustered motif arrangements outperform interleaved arrangements in all 8 paired conditions, by a mean of +0.129 (SD 0.086, 95% CI +0.057 to +0.201; paired t on condition means, $t(7) = 4.23$, $p = 0.0039$; maximum +0.294). Two of the eight remain significant individu-

ally after Benjamini-Hochberg correction. The mean exceeds the gain from adding a second motif family, so grammar rather than content is the larger available lever. Two caveats: the eight conditions share backgrounds and reuse motif pairs across loads, so they are not fully independent and the p-value is optimistic; and Wilcoxon and sign tests both return 0.0078, the resolution floor at $n = 8$.

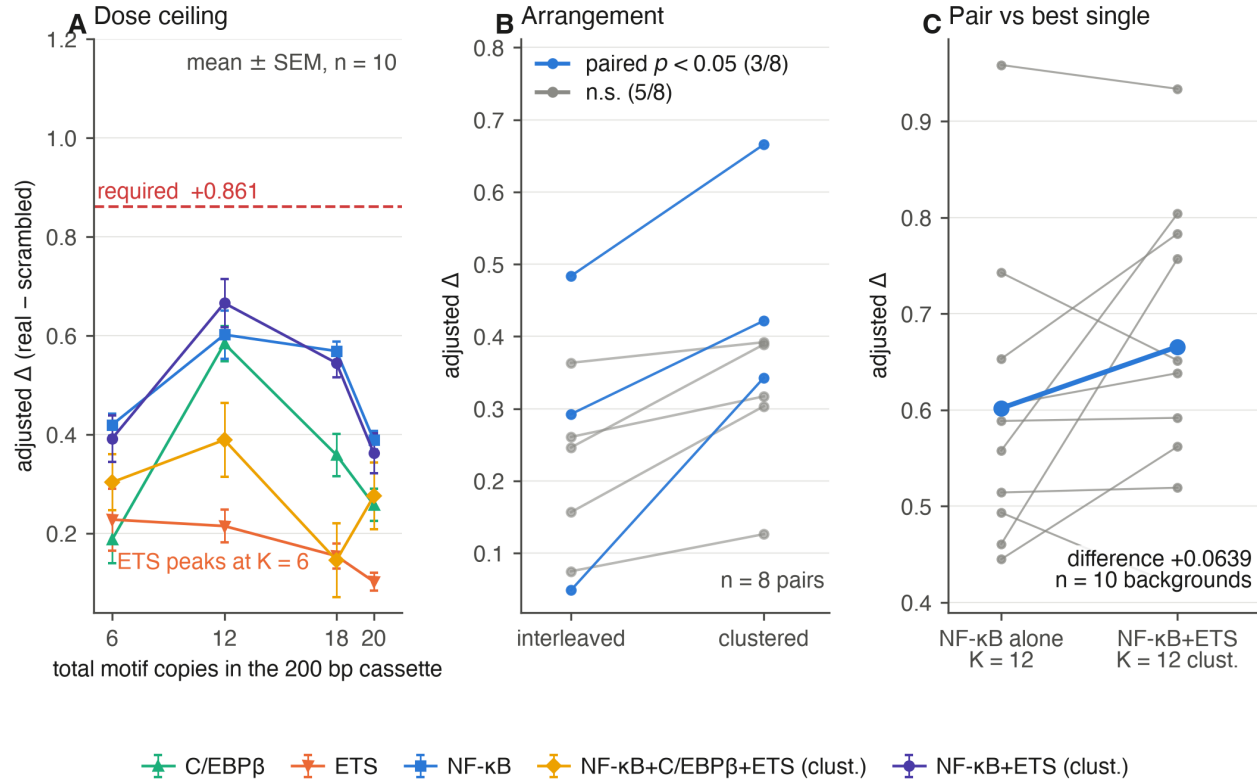


Figure 6. The design ceiling. Predicted activity gain from motif insertion into random 200 bp background. Panels show dose response against motif copy number with the +0.861 requirement marked, where four of five conditions peak at 12 copies and ETS peaks at 6, and all five fall below their peak at 20; clustered against interleaved arrangements paired across the eight conditions; and the best pair against the best single family at matched insertion load, paired by background. Error bars are SEM over $n = 10$ random backgrounds, as labelled on the panels.

3.8 Insertion disagrees with observation across most motif families

AP-1 insertion into random background lowers predicted senescence H3K27ac: delta -0.464, monotonic, Spearman -0.375, against a scrambled control of -0.016. This contradicts AP-1's established role as a pioneer factor imprinting the OIS enhancer landscape.

Ablation resolves it. Removing native AP-1 sites from real regions gives -0.1055 ($p = 0.032$) in high-response regions but +0.0658 in low-response regions, interaction $p = 0.0091$, continuous Spearman -0.364 ($p = 0.011$, $n = 48$). An NF- κ B positive control behaves as expected (-0.1404, $p = 0.0003$), validating the assay. AP-1 is context-dependent: its presence supports activity where activity is already high, and insertion into a weak background does not.

Observational data agrees. AP-1 motif frequency rises monotonically across response deciles, from 0.086 to 0.179 (OR 2.31, $p = 5.9e-94$). AP-1 regions carry +0.206 higher H3K27ac ($p = 2e-283$, $n = 16,909$). AP-1 shows the highest in-silico mutagenesis attribution of the four families tested (3.61-fold, $p = 0.0002$, $n = 16$).

The alternative explanation — that AP-1 opens chromatin while depleting flanking acetylation — fails in the opposite direction: even within the top accessibility quartile, AP-1 still predicts higher H3K27ac ($p = 4.9\text{e-}05$).

The generalizable finding is that this discordance is not specific to AP-1. Three of four motif families show it: C/EBP-beta and ETS are depleted in high-H3K27ac regions yet produce positive insertion deltas, and AP-1 shows the reverse. What a model predicts for a synthetic insertion and what it learned from genomic context are systematically different quantities. Any design pipeline that optimizes the former while validating against the latter is optimizing something other than what it measures.

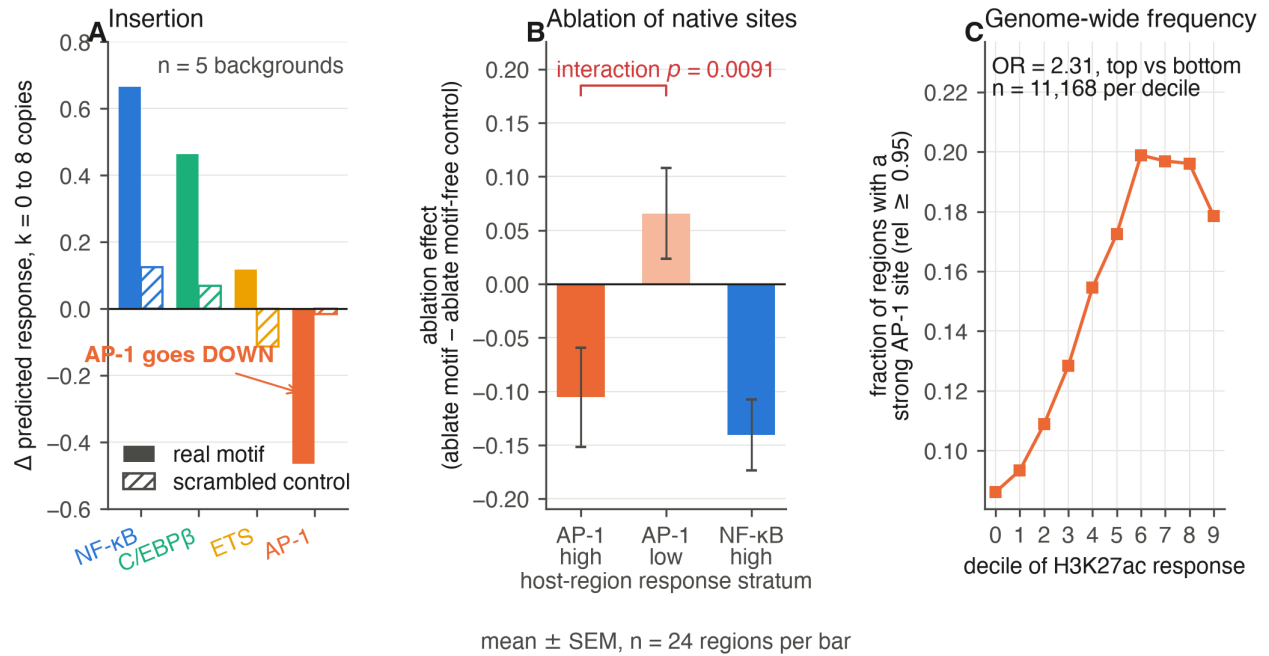


Figure 7. AP-1 context dependence. Panels show the predicted effect of inserting each motif family into random background with matched scrambled controls; the effect of ablating native sites from real regions, stratified by whether the region is high- or low-response, with NF- κ B as a positive control and the interaction p stated; and AP-1 motif frequency across response deciles. Insertion and ablation point in opposite directions for AP-1, which is the finding: AP-1 supports activity where activity is already high and does not create it in a weak background. Error bars are SEM over $n = 24$ regions per bar in the ablation panel, as labelled.

4. Discussion

Replicative senescence is the canonical model. It is where the field began, it is what most introductory accounts describe, and a substantial body of senescence epigenomics rests on it. We find that its enhancer-level response does not reproduce across independent datasets, including between two datasets deposited by the same laboratory through its own single pipeline. The same measurement applied to oncogene-induced senescence returns +0.653 across far greater technical distance.

Two comparisons bound the alternative explanations. The ceiling itself is cross-laboratory, so the contrast is not between a within-study and a between-study measurement: two laboratories measuring oncogene-induced senescence seven years apart agree at +0.653, while one laboratory measuring replicative senescence in two cell lines agrees at +0.054, rising to +0.146 on regions shared between the two comparisons against +0.645 for the ceiling under the same restriction. And holding laboratory, genome build and cell line

all constant while varying only mechanism gives +0.114, so neither laboratory identity nor cell line can carry the replicative result. The practical implication is narrow and concrete. Computational work that treats published replicative-senescence enhancer landscapes as a stable target — motif discovery, regulatory network inference, model training — may be fitting dataset-specific variation. We are not claiming that replicative senescence is not a real cell state; the CDKN2A promoter response is significantly negative within each replicative arm individually, which argues that it is. We are claiming that its distal regulatory programme, as captured by current public data, does not replicate.

Why it might not is beyond what these data can settle. One possibility is biological: replicative senescence is reached gradually over many divisions, each culture accumulating its own history of damage and drift, so cultures arriving at the arrest may have taken different regulatory paths, where oncogene-induced senescence is a synchronized response to a defined stimulus. A second is statistical: every replicative dataset here provides $n = 2$ per arm, and a weak but real programme might be undetectable at that replication. The same-laboratory comparison argues against the second explanation without eliminating it.

Relation to prior work

Senescence heterogeneity across inducers is well established, and several studies reach directionally consistent conclusions by other routes.

Bridge et al. (2026) drive IMR-90 to senescence by four inducers under matched within-laboratory conditions and report that global transcriptomic variation aligns along a shared axis of senescence progression, that gene-level overlap is limited with most responses inducer-specific, and that pathway-level enrichment is far more consistent. Their hierarchical model — conserved pathways, divergent genes — is compatible with what we observe, and our expression-level measurement (+0.531 between two OIS studies against +0.142 and +0.093 between replicative and either) is the cross-dataset version of their within-laboratory comparison. Three differences matter. Theirs is transcriptome and ours H3K27ac, so a shared pathway-level response need not imply a shared enhancer-level one. Theirs is within one laboratory and cell line, which removes the technical variation we measure across. And theirs constructs no within-mechanism positive control, so there is no reference for what agreement looks like when the mechanism is shared — the comparison that makes a low value interpretable rather than merely low.

Casella et al. (2019) take the complementary approach, searching eight senescence models for a shared core and recovering 47 consistently elevated and 26 consistently reduced transcripts. A small conserved core and a non-reproducing enhancer programme are not in conflict: a handful of genes can move consistently while the elements driving the bulk of the response do not.

At single-cell resolution, replicative senescence populations show broad transcriptomic heterogeneity while irradiation- and etoposide-induced populations resemble each other more closely (Wechter et al. 2023, in WI-38) — consistent, by a different method, with replicative senescence being the mechanism whose response fails to reproduce.

Sati et al. (2020) compare OIS and replicative senescence at the level of 3D genome architecture and find them markedly different, OIS gaining compartmentalization and SAHFs where replicative senescence loses it. Our result is the cis-regulatory counterpart.

Most closely on method, Yuan et al. (2026) fine-tune a frozen Borzoi trunk onto replicative-senescence data and report $R = 0.49$ for held-out expression change, with promoter-motif enrichment at only $R = 0.15$, which they attribute to senescence-induced genes being enhancer-driven. We arrive at the same conclusion about promoters from a different direction, and our results suggest an additional reason their promoter signal is weak: the replicative-senescence target itself does not reproduce across datasets.

For model-guided design of state-specific elements

The mechanism must be chosen before the design target. Pooling training data across mechanisms is harmless and can stabilize a null; pooling the prediction target is not, costing 0.095 AUC and a control in our hands.

A model that classifies well may still be unable to guide design. Ours reaches 86% of the data's own reproducibility ceiling and 0.907 AUC on a cross-laboratory holdout, yet the strongest available motif lever reaches only 0.773 of what moving a sequence to the 95th percentile requires, and additional motif content past the peak reduces predicted activity. Reporting classification performance as evidence of design readiness conflates two capabilities.

The insertion-versus-observation discordance deserves attention beyond this system. In three of four motif families, what the model predicts for synthetic insertion disagrees with what it learned about that motif from genomic context. Design pipelines routinely optimize the former and interpret the result through the latter.

5. Limitations and unresolved gaps

1. **n = 2 per arm throughout.** Every human dataset used here provides two replicates per arm; none reaches three. A within-replicative comparison at n of three or more on both sides would settle whether the non-reproducibility is biological or statistical, and no such pair exists in public data.
2. **All results are computational.** No designed element was synthesized or tested. The design ceiling is a property of the model's predictions rather than a measured property of DNA.
3. **No dataset was reprocessed from raw reads.** All analyses read deposited bigWig signal, and for GSE175533 a deposited log2 fold change that was not requantified. Claims of a shared processing pipeline rest on the depositing laboratory's own pipeline, evidenced by a common filename convention, rather than on uniform requantification here.
4. **The OIS model rests on two laboratories and two cell lines.** A third independent OIS dataset exists but is deposited as raw reads only and was not processed.
5. **The load-bearing comparison retains a biological confound.** The comparison with no technical confound pairs two different cell lines, IMR-90 against BJ. Comparison X1 bounds the cell-line contribution but does not eliminate it.
6. **The distal CDKN2A pattern rests on one dataset.** Two of three replicative datasets are positive at distal regions, the same sign as both OIS datasets. With two OIS against three replicative arms, no test can distinguish the groups; the pattern is reported descriptively.
7. **Therapy-induced senescence has no public chromatin dataset** with matched controls and replicates. Our results give no basis for assuming transfer into it from either mechanism studied here.
8. **Correlations rest on different region sets.** Recomputing on a common set leaves the ordering intact and the ceiling essentially unchanged, but moves the smallest within-replicative value from +0.054 to +0.146. Both are reported wherever that value is quoted.

What would change the conclusion

A senescent-cell massively parallel reporter assay would permit training on measured regulatory activity rather than on accessibility or acetylation as proxies, and would be the single highest-value addition to this field's data.

A replicative-senescence chromatin dataset at n of 3 or more per arm, compared against an independent dataset of comparable power, would test the statistical explanation directly.

Therapy-induced senescence chromatin data with matched proliferating controls would establish whether a

third mechanism also diverges, and would make the most experimentally tractable senescence model available to sequence modelling.

Data and Code Availability

All data used here is public. Accessions: GSE175533, GSE146585, GSE106146 (within GSE105937), GSE206402 and GSE205898 (within GSE206496), GSE74238 (within GSE74328), GSE74324, PRJNA439280, GSE210285 (excluded), GSE115940, GSE99028, GSE157867, GSE117210, GSE220545, GSE205692, and ENCODE experiments ENCSR200OML, ENCSR000CTK, ENCSR000CTQ. A complete verification log recording the method and date of live verification for all 44 accessions touched by this work accompanies the deposition.

Analysis scripts, figures with stand-alone captions, result tables, the full numerical audit trail and the environment record are available at <https://github.com/gabeykim/senescence-enhancer-reproducibility>. Processed region-response tables, the harmonized training sets, trunk embedding caches and per-region importance arrays are deposited at <https://doi.org/10.5281/zenodo.23146107>, allowing every downstream result to be reproduced without GPU access, since the expensive step is cached. Raw sequencing data is not redeposited; a fetch script reproduces it from the original accessions.

One limitation of the environment record should be stated directly. The GPU installer was run without capturing a transitive dependency freeze, and those instances are no longer available, so a fully resolved GPU lockfile cannot be reconstructed. The deposited record marks each dependency as evidenced, inferred, or not captured rather than asserting pins that were never verified. The local analysis environment, covering the numerical audit and all figure generation, is exact.

The repository retains the working analysis reports alongside the final values, with differences between them documented.

Several software issues encountered during this work have been reported upstream to gReLU: the aggregation utility requires an explicit length aggregation function or fails on Borzoi's bin count; a sequence-return option raises a key error; a checkpoint flag raises a spurious type error through an identity comparison; and a documented environment variable for genome paths is not read.

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Competing Interests

The author declares no competing interests.

Author Contributions

G.K. conceived the study, performed all analyses, and wrote the manuscript.

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Every citation was verified against a canonical publisher record.

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